Code No: 234AG

KR23

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025

COMPUTER ORGANIZATION AND ARCHITECTURE

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) Define Computer Architecture. [1M] L1
   b) Draw the block diagram of digital computer with components. [1M] L1
   c) Name the most common fields found in instruction format. [1M] L1
   d) What is the importance of addressing mode in an instruction. [1M] L2
   e) Find the 2's complement of the binary number 10101. [1M] L2
   f) Convert decimal number 53 into binary system. [1M] L2
   g) Explain the purpose of ROM portion of main memory. [1M] L2
   h) Define Hit Ratio. [1M] L1
   i) Demonstrate about parallel processing. [1M] L2
   j) List the RISC characteristics. [1M] L1

PART- B (5 X 10 =50 Marks)

2. a) Draw and explain the flowchart for interrupt cycle in detail. [7M] L2
   b) Explain about the registers for basic computer with the storage size. [3M] L2

OR

3. a) Discuss about Computer instructions in detail. [7M] L2
   b) Write the differences between computer organization and architecture. [3M] L2

4. List and explain about different addressing modes in detail with examples. [10M] L2

OR

5. a) Explain the three address instruction format representation with an example. [5M] L3
   b) Discuss about data manipulation instructions. [5M] L2

6. a) Explain different types of data representations used in digital computer with examples. [5M] L.

b) Discuss about fixed-point representation and its limitations. [5M] L2

OR

7. a) Explain the importance of normalization in floating point arithmetic with an example. [5M] L2

b) Compare signed magnitude, 1's Complement and 2's Complement representations. [5M] L3

8. a) Illustrate the components of typical memory hierarchy. [5M] L3

b) Compare Static RAM and Dynamic RAM. [5M] L2

OR

9. Illustrate about the main memory connection to CPU with the help of a diagram. [10M] L2

10. a) Explain about arithmetic pipeline in detail. [5M] L2

b) Demonstrate about instruction pipeline flowchart. [5M] L2

OR

11. a) Distinguish between static branch prediction and dynamic branch prediction. [5M] L4

b) Summarize the control hazards in pipelining. [5M] L2

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B.Tech II Year II Semester Regular Examinations, June/July-2025

COMPUTER ORGANIZATION AND MICROPROCESSOR

(INFORMATION TECHNOLOGY)

Time: 3 hours Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a)	Define an interrupt. List the type interrupts.	[1M] L2
b)	What are memory reference instructions? Give an example.	[1M] L2
c)	Define Instruction.	[1M] L1
d)	List the various addressing modes present in 8086.	[1M] L2
e)	What is the function of passing parameters to PROCEDURES in 8086?	[1M] L2
f)	What is the purpose of the stack in 8086?	[1M] L2
g)	Name any two modes of data transfer.	[1M] L1
h)	What is the main function of an 8089 IOP?	[1M] L1
i)	Define cache memory.	[1M] L1
j)	What is pipelining in computer architecture?	[1M] L2

PART- B (5 X 10 =50 Marks)

2. a)	With Neat Sketch explain Computer Architecture.	[5M] L2
b)	Explain different types of computer instructions with examples.	[5M] L3
	OR	
3.	Explain BUS Architecture in Computer Organization.	[10M] L2
4. a)	Explain Physical memory organization of 8086.	[5M] L2
b)	Explain Special Processor Activities in 8086.	[5M] L2
	OR	
5. a)	Discuss the general BUS organization of the 8086 microprocessor.	[5M] L2
b)	Explain with neat sketch I/O addressing of 8086.	[5M] L2

6.	a) Explain Timing diagram of 8086.	[5M] L3
	b) Write an ALP to move 10 blocks of data from one memory to another.	[5M] L2
	OR	
7.	a) What is the role of Assembler Directives in 8086 programming? Explain any five.	[5M] L3
	b) Write a short note on Macros in 8086.	[5M] L2
8.	Discuss the architecture and operation of Direct Memory Access (DMA) for 8086.	[10M] L3
	OR	
9.	a) Describe the working of an Asynchronous Data Transfer in 8086.	[5M] L2
	b) Write a short note on Priority Interrupt and its types.	[5M] L3
10.	a) Differentiate between Main Memory and Auxiliary Memory.	[5M] L4
	b) What is Instruction Pipelining? Give an example.	[5M] L3
	OR	
11.	Discuss the working of Array Processors in detail.	[10M] L4

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Code No: 234AJ

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Kmit

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech II Year II Semester Regular Examinations, June/July-2025

COMPUTER ORGANIZATION AND ARCHITECTURE

CSE (AI&ML)

Time: 3 hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (10 Marks)

1. a) What is a micro operation? Give two examples. [1M] L2
   b) What is the function of the control unit in a CPU? [1M] L2
   c) Define Rotate Left and Rotate Right operations. [1M] L1
   d) Mention the difference between direct and indirect addressing modes. [1M] L2
   e) What is the purpose of the complement system in binary arithmetic? [1M] L2
   f) Find the 9's complement of 765. [1M] L2
   g) Define DMA and mention two advantages of using it. [1M] L1
   h) What is Associative Memory? How is it different from normal RAM? [1M] L2
   i) Define cache coherence. [1M] L1
   j) Define mutual exclusion and semaphore. [1M] L1

PART- B (5 X 10 =50 Marks)

2. a) Discuss the design and working of a 4-bit binary adder with a neat diagram. [5M] L3
   b) Discuss the design and working of a 4-bit binary adder-subtractor with a neat diagram. [5M] L3

OR

3. Explain in detail about interrupt cycle with a neat flowchart. [10M] L2

4. Explain the concept of control memory in micro programmed control. Describe how it differs from main memory and its role in instruction execution [10M] L2

OR

5. Explain the different types of addressing modes with examples. [10M] L3

6. a) Explain the steps involved in floating point addition and subtraction with a flowchart. [6M] L3

b) Differentiate between 1's compliment a d 2's complement representations with an example. [4M] L3

OR

7. Perform 23X19 by using signed magnitude multiplication algorithm with a flowchart. [10M] L3

8. a) Explain the architecture and working of an Input Output interface. [7M] L2

b) Describe the methods of Asynchronous Data Transfer. [3M] L2

OR

9. a) Describe in detail the working of Cache Memory. What are the cache mapping techniques, explain in detail with relevant diagrams. [8M] L3

b) Describe the concept of Memory Hierarchy. [2M] L3

10. a) Describe the organization and working of a RISC pipeline. [6M] L3

b) Explain the concepts of delayed load and delayed branch with examples. [4M] L3

OR

11. What is inter processor arbitration? Explain static and dynamic arbitration techniques with diagrams. [10M] L2

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B.Tech II Year II Semester Regular Examinations, June/July-2025

COMPUTER ORGANIZATION AND ARCHITECTURE

CSE (DATA SCIENCE)

Time: 3 hours

Max. Marks: 60

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 10 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

(10 Marks)

1. a) What are the needs of studying Computer Organization. [1M] L1
   b) Demonstrate briefly about computer register. [1M] L2
   c) List the types of CPU organization. [1M] L1
   d) Explain the role of operand in an instruction code with an example. [1M] L2
   e) Illustrate about overflow in arithmetic. [1M] L2
   f) Define fixed point representation. [1M] L1
   g) Outline the disadvantage of strobe method? [1M] L2
   h) Define Cycle Stealing. [1M] L1
   i) List CISC characteristics. [1M] L1
   j) What is pipelining? [1M] L1

PART- B

(5 X 10 =50 Marks)

2. Explain the memory reference instructions in detail with the help of flow chart. [10 M] L2

OR

3. a) Illustrate about common bus system with a neat diagram. [7M] L2
   b) Identify the list of registers for basic computer with the storage size. [3M] L2

4. Explain about instruction format and its types with suitable examples. [10M] L2

OR

5. a) Discuss about general register organization in detail. [5M] L2
   b) Explain about data transfer instructions. [5M] L2

6. a) Describe 1's & 2's complement methods for representing negative numbers with examples. [5M] L2

b) Convert -32 to its 8-bit 2's complement form detailing the step-by-step procedure. [5M] L3

OR

7. Discuss the algorithm and hardware implementation for the addition and subtraction of signed magnitude data with flowchart and examples. [10M] L3

8. a) Explain RAM and ROM with the help of block diagrams. [5M] L2

b) Discuss about typical DMA controller with the help of block diagram. [5M] L2

OR

9. a) Compare Cache memory and Main memory in detail. [5M] L2

b) Explain Daisy-Chaining priority interrupt in detail. [5M] L2

10. Distinguish between RISC and CISC in detail. [10M] L4

OR

11. a) Illustrate about hazards associated with pipeline, in detail. [5M] L2

b) Explain about Flynn's classification. [5M] L2

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RKR21

Code No: 223AB

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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B.Tech II Year I Semester Supplementary Examinations, July-2025

COMPUTER ORGANIZATION AND ARCHITECTURE

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (20 Marks)

1. a) Define computer architecture. How is it different from computer organization? [2M] L2
   b) Differentiate between opcode and operand. [2M] L1
   c) What is the role of general-purpose registers in a CPU? [2M] L1
   d) What is Instruction Set Architecture (ISA)? [2M] L1
   e) What is a priority interrupt? [2M] L2
   f) Differentiate between main memory and auxiliary memory. [2M] L2
   g) What is parallel processing? [2M] L1
   h) What does RISC stand for? Mention one characteristic. [2M] L1
   i) What is data parallelism? Give an example. [2M] L2
   j) What is High Performance Computing (HPC)? [2M] L2

PART- B (5 X 10 =50 Marks)

2. a) Explain the block diagram of a digital computer and describe its major components. [5M] L2
   b) Describe the instruction cycle in detail with the help of a diagram. [5M] L2

OR

3. Explain the different types of computer instructions: memory-reference, register-reference, and input-output instructions. [10M] L2

4. a) Explain various addressing modes with suitable examples. [5M] L2
   b) Describe the steps involved in binary multiplication using Booth's Algorithm. [5M] L2

OR

5. a) Explain general register organization with a neat diagram. How does it improve instruction execution? [5M] L3

b) Explain the process of binary addition and subtraction with overflow detection. [5M] L2

6. a) Explain the memory hierarchy in computer systems with a neat diagram. [5M] L2

b) Explain Direct Memory Access (DMA) and its working with a diagram. [5M] L2

OR

7. Describe the modes of data transfer: Programmed I/O, Interrupt-Driven I/O, and DMA. [10M] L2

8. a) Describe the working of arithmetic and instruction pipelines. How are they different? [5M] L2

b) Discuss the major design issues in pipelining and their impact on performance. [5M] L3

OR

9. a) Compare the characteristics of RISC and CISC architectures in detail. [5M] L2

b) Explain the three types of pipeline hazards: structural, data, and control. Give suitable examples. [5M] L2

10. a) Compare CPU and GPU architectures. How do GPUs achieve high performance? [5M] L3

b) Explain the goals and benefits of parallel computing. [5M] L2

OR

11. a) Explain matrix multiplication using CUDA. [5M] L3

b) Explain the structure of a CUDA program with an example. [5M] L3

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Code No: 223AB **RKR21**

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B.Tech II Year I Semester Regular Examinations, February-2024

Computer Organization and Architecture

(Common to IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (2 X 10 =20 Marks)

1. a) Why do the computer system used a common bus structure? [2M] L2
   b) Write the instruction format for memory reference and register reference instruction [2M] L2
   c) What is the significance of Accumulator in instruction execution? [2M] L2
   d) Draw the flowchart to perform subtraction of 2's complement data. [2M] L2
   e) Differentiate between write through and write back policy in cache updation. [2M] L2
   f) What is the need of I/O Interface in a computer system? [2M] L2
   g) Why do we need to implement a system with arithmetic pipeline? [2M] L2
   h) Summarize any four characteristics of RISC processor architecture. [2M] L2
   i) Identify any four goals of parallel computing. [2M] L2
   j) Outline the steps involved in CUDA programming structure. [2M] L2

PART- B (5 X 10 =50 Marks)

2. a) How do the interrupt is handled during the execution of an instruction. Demonstrate with a neat diagram. [5M] L3
   b) Identify any five Input-Output instructions in a basic computer system and illustrate with an example for each. [5M] L3

OR

3. a) What are the three types of instruction code formats? Explain the instruction format for each type of instruction. [5M] L3
   b) What is a micro operation? What are the sequence of micro-operations for ADD and BUN memory reference instructions [5M] L3

4. a)	How do you represent two address instruction format? Explain with an example to evaluate an arithmetic expression x= (A+B)*(C+D).	[5M] L3
b)	Illustrate with an example the purpose of immediate addressing and relative addressing modes.	[5M] L3
	OR	
5. a)	Describe with a flowchart how to perform multiplication of two numbers using Two's complement data. (Booth Multiplication algoithm)	[5M] L3
b)	Write the sequence of steps to multiply +9 and +13 using Booth's Multiplication algoithm	[5M] L3
6. a)	What is cache memory? Illustrate with a neat sketch how associative mapping is implemented.	[5M] L3
b)	What is data transfer? Describe how DMA is used for efficient data transfer.	[5M] L3
	OR	
7. a)	Describe any two auxiliary storage devices used in computer system.	[5M] L2
b)	Compare synchronous and asynchronous modes of data transfer.	[5M] L2
8. a)	How does Flynn's classify computer architectures? Demonstrate with a neat sketch MISD architecture.	[5M] L3
b)	What are structural hazards in instruction pipelining? How these hazards are addressed during execution.	[5M] L3
	OR	
9. a)	Compare static and dynamic branch prediction mechanisms in instruction pipeline.	[5M] L4
b)	What is RISC pipeline? Illustrate with a neat sketch how it is implemented.	[5M] L2
10. a)	Describe the differences between CPU and GPU architectures.	[5M] L2
b)	Outline the applications of data parallelism in high performance computing.	[5M] L3
	OR	
11. a)	Describe the various CUDA memory types.	[5M] L2
b)	Explain about CUDA kernel functions and threading.	[5M] L2

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B.Tech II Year I Semester Regular Examinations, February-2024

Computer Organization and Architecture

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A **(2 X 10 =20 Marks)**

1. a) Why do a computer system need Random access memory? [2M] L1
   b) How do you compare direct address and indirect address with an example? [2M] L1
   c) What is the significance of addressing mode in an instruction? [2M] L1
   d) Identify the fields in representing a floating point number. [2M] L1
   e) Differentiate between static RAM and Dynamic RAM. [2M] L2
   f) Summarize the various modes of data transfer between I/O and Memory. [2M] L2
   g) Why does a computer system need parallel processing? [2M] L1
   h) Identify the various hazards that arise in instruction pipelining. [2M] L2
   i) Differentiate between CPU and GPU. [2M] L2
   j) Summarize any four applications of parallel computing. [2M] L2

PART- B **(5 X 10 =50 Marks)**

2. a) Demonstrate with a flowchart the various stages of instruction cycle. [5M] L3
   b) Identify any five memory reference instructions and explain the purpose of each. [5M] L3

OR

3. a) What are the basic components of a computer system? Explain the need of control unit in a computer with a neat sketch. [5M] L2
   b) What is a micro operation? What are the sequence of micro-operations for ADD and STA memory reference instructions [5M] L3

4. a) How do you represent three address instruction format? Explain with an example to evaluate an arithmetic expression x= (A+B)*(C+D). [5M] L3
   b) Illustrate with an example the purpose of direct and indirect addressing modes. [5M] L3

OR

5. a) Describe with a flowchart how to perform multiplication of two numbers using signed magnitude data. [5M] L3
   b) Write the sequence of steps to multiply 8 and 9 using signed magnitude algorithm. [5M] L3

6. a) What is cache memory? Illustrate with a neat sketch how set associative mapping cache is implemented. [5M] L3

   b) What is data transfer? Describe how Programmed I/O is used for data transfer. [5M] L3

OR

7. a) Demonstrate with a neat sketch how virtual memory is implemented. [5M] L2

   b) What is strobe protocol? Explain how to perform data transfer using it. [5M] L2

8. a) How does Flynn's classify computer architectures? Demonstrate with a neat sketch MIMD architecture. [5M] L3

   b) What are control hazards in instruction pipelining? How the control hazards are resolved during execution. [5M] L3

OR

9. a) What are the characteristics of a RISC processor? What are the advantages of it over CISC processor [5M] L4

   b) What is an instruction pipeline? Illustrate with a neat sketch how it is implemented. [5M] L2

10. a) What is parallel computing? Explain the architecture of modern GPU. [5M] L2

    b) Describe the CUDA programming structure. [5M] L2

OR

11. a) Illustrate with a neat sketch CUDA memory hierarchy [5M] L3

    b) Explain about device global memory and data transfer. [5M] L2

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COMPUTER ORGANIZATION AND ARCHITECTURE

(COMMON TO CSE, IT, CSE(AI&ML), CSE(DS))

Time: 3 hours Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (20 Marks)

1. a) Define computer architecture. How is it different from computer organization? [2M] L2
   b) Differentiate between opcode and operand. [2M] L1
   c) What is the role of general-purpose registers in a CPU? [2M] L1
   d) What is Instruction Set Architecture (ISA)? [2M] L1
   e) What is a priority interrupt? [2M] L2
   f) Differentiate between main memory and auxiliary memory. [2M] L2
   g) What is parallel processing? [2M] L1
   h) What does RISC stand for? Mention one characteristic. [2M] L1
   i) What is data parallelism? Give an example. [2M] L2
   j) What is High Performance Computing (HPC)? [2M] L2

PART- B (5 X 10 =50 Marks)

2. a) Explain the block diagram of a digital computer and describe its major components. [5M] L2
   b) Describe the instruction cycle in detail with the help of a diagram. [5M] L2

OR

3. Explain the different types of computer instructions: memory-reference, register-reference, and input-output instructions. [10M] L2

4. a) Explain various addressing modes with suitable examples. [5M] L2
   b) Describe the steps involved in binary multiplication using Booth's Algorithm. [5M] L2

OR

5\. a) Explain general register organization with a neat diagram. How does it improve instruction execution? [5M] L3

b) Explain the process of binary addition and subtraction with overflow detection. [5M] L2

6\. a) Explain the memory hierarchy in computer systems with a neat diagram. [5M] L2

b) Explain Direct Memory Access (DMA) and its working with a diagram. [5M] L2

OR

7\. Describe the modes of data transfer: Programmed I/O, Interrupt-Driven I/O, and DMA. [10M] L2

8\. a) Describe the working of arithmetic and instruction pipelines. How are they different? [5M] L2

b) Discuss the major design issues in pipelining and their impact on performance. [5M] L3

OR

9\. a) Compare the characteristics of RISC and CISC architectures in detail. [5M] L2

b) Explain the three types of pipeline hazards: structural, data, and control. Give suitable examples. [5M] L2

10\. a) Compare CPU and GPU architectures. How do GPUs achieve high performance? [5M] L3

b) Explain the goals and benefits of parallel computing. [5M] L2

OR

11\. a) Explain matrix multiplication using CUDA. [5M] L3

b) Explain the structure of a CUDA program with an example. [5M] L3

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B.Tech II Year I Semester Supplementary Examinations, February-2024

Computer Organization and Architecture

(Common to CSE, IT, CSE (AI&ML), CSE (DATA SCIENCE))

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (2 X 10 =20 Marks)

1. a) Define computer organization. [2M]L1
   b) What is meant by an instruction cycle? [2M]L1
   c) What is an instruction code. [2M]L1
   d) Differentiate between hard wired program control unit and microprogram control unit [2M]L1
   e) Explain the following secondary storage devices a) Magnetic disk. b) Magnetic tape [2M]L2
   f) Define Virtual Memory? [2M]L1
   g) What is dynamic branch prediction? [2M]L1
   h) What are the steps follow in athematic pipeline [2M]L1
   i) What is parallel computing? [2M]L1
   j) What do you mean by CUDA core? [2M]L1

PART- B (5 X 10 =50 Marks)

2. a) Explain Instruction cycle [5M] L2
   b) Draw and explain functional unit of basic computer system [5M] L2

OR

3. a) List Memory Reference Instruction and explain it. [5M] L2
   b) Give list of basic computer register with their size and draw register organization with memory bank [5M] L3

4. a) Describe different kinds of addressing modes with an example [5M] L3
   b) Differentiate arithmetic shift and logical shift [5M] L4

OR

5. a) Subtract $(11010)_2$-$(10000)_2$ using 1's complement and 2's compliment method [5M] L3
   b) Write down the booth's algorithm? List the features of booth's algorithm [5M] L2

6. a) In a hierarchical memory system, where does the cache memory placed ? Explain the terms 'locality of reference' and 'cache line' [5M] L2

b) Distinguish between the strobed I/O and interrupt driven data transfer modes [5M] L4

OR

7. a) Explain cache memory with any one mapping technique [5M] L2

b) Explain Associative Memory [5M] L2

8. a) Explain characteristics of RISC and CISC. [5M] L2

b) Explain in detail about Parallel Processing [5M] L2

OR

9. a) Explain in detail about four-segment instruction pipeline [5M] L3

b) What are the difference between Data Hazards and control Hazards [5M] L2

10. a) What are the goals of parallel computing [5M] L2

b) What is the difference between CPU & High Performance computing [5M] L2

OR

11. a) What is the use of cudaMalloc(),cudaMemcpy() function and what arguments does it accept? [5M] L2

b) Which properties of GPU are used to find its compute capability? How to set any GPU as current GPU when its id is given? [5M] L2

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